OmniDesk AI — Technical Specification & A-Z Guide

Enterprise AI-Powered Customer Support, SLA Management & Automated Ticket Escalation Desk

Author: Md. Asif Iqbal | **System Architecture:** Next.js 16 + Node.js Express + MongoDB Atlas + Gemini/Groq AI

1. Executive Summary: What Does This Project Do?

OmniDesk AI is an intelligent, high-concurrency customer support desk designed to eliminate ticket backlog, automate SLA breach prevention, and dramatically decrease customer churn. Unlike legacy ticketing platforms (such as basic Jira or Zendesk) where agents manually review and assign tickets, OmniDesk AI introduces a multi-tier automation workflow driven by large language models (LLMs) and real-time SLA calculation engines.

- **Real-Time NLP Triage:** As a customer writes their ticket, multimodal AI (Google Gemini 3.8 Flash + Groq) evaluates the text in milliseconds, classifying category and priority while predicting customer distress.
- **Algorithmic SLA Enforcement:** Tickets are governed by strict resolution deadlines (Urgent: 2 hours, High: 8 hours, Medium: 24 hours, Low: 48 hours) with real-time countdowns.
- **Autonomous Ticket Escalation:** If a customer expresses acute frustration (Sentiment Score ≤ -0.6) or if an urgent ticket remains unacknowledged for over 50% of its SLA window, the system autonomously triggers Tier-2 escalation.
- **AI Agent Copilot:** Generates instant, context-aware suggested responses with 1-click clipboard and reply-box insertion, reducing First Response Time (FRT) by over 75%.
- **Granular Role-Based Access Control (RBAC):** Isolated and secure operational views for Customers, Support Agents, and System Administrators.

2. Full-Stack System Architecture & Technology Stack

Frontend Application	Next.js 16 (App Router), React 19, Tailwind CSS v4, Lucide Icons	Turbopack-powered responsive interface (320px–1440px+). Zero layout shifts, custom theme palette, live SLA badges.
Authentication & Identity	Firebase Auth + Custom Node.js JWT (HMAC-SHA512)	Google Sign-In Popup & Email/Password synced to MongoDB with 512-bit signed authorization bearer tokens.
Backend REST API	Node.js, Express.js, TypeScript, Zod Schema Validation	Strict MVC controllers, robust error middleware, CORS credential handling, and automated SLA breach cron engine.
AI & NLP Engine	Google Gemini 3.8 Flash + Groq (GPT-OSS-20B) + Heuristic Fallback	Multi-model orchestration: Groq for millisecond draft replies, Gemini for multimodal triage, heuristic regex for offline safety.
Persistence Layer	MongoDB Atlas (Cloud Cluster M0 / Replica Set)	Mongoose ODM models for Tickets, Users, Internal Comments, and immutable Escalation Audit Logs.

3. Role-Based Workflows (Customer, Agent, Admin)

Role	Dashboard View	Core Capabilities & Operational Actions
Customer	Personal Ticket Portal	Creates support tickets with real-time AI preview; reviews ticket status, SLA timer, and conversation responses from agents.
Support Agent	OmniDesk Triage Desk	Claims tickets ('Assign to Me'), sends public responses, logs private 'Internal Notes' (hidden from customers), uses AI Suggested Replies, and marks tickets as Resolved.
Administrator	Executive Analytics Hub	Monitors company-wide SLA breach rates, active queues, re-assigns tickets across staff, and audits the complete escalation timeline.

4. End-to-End Ticket Lifecycle (A-to-Z Execution Stages)

Stage 1: Ticket Ingestion & Live AI Evaluation

The customer enters problem details. As they type, client-side debouncing triggers AI evaluation. Gemini/Groq evaluates text sentiment, urgency score (0–100), and automatically recommends appropriate Priority and Category.

Stage 2: Deterministic SLA Calculation & Atlas Persistence

Upon ticket submission, the backend SLA engine computes an immutable deadline: SLA Deadline = CreatedAt + SLA Hours Map. Ticket status is set to 'OPEN' and stored in MongoDB Atlas with initial AI audit metrics.

Stage 3: Queue Triage & Ticket Claim

Support agents view tickets sorted by urgency and SLA risk. Clicking 'Assign to Me' claims the ticket, transitioning status from 'OPEN' to 'IN_PROGRESS' and updating timestamps.

Stage 4: AI Copilot Response Generation & Internal Notes

The AI Triage Copilot sidebar provides a pre-drafted, empathetic response. The agent reviews it, clicks 'Use Draft in Reply', and dispatches it to the customer. For sensitive investigations, agents write 'Internal Notes' visible only to internal staff.

Stage 5: Autonomous or Manual Escalation

If an SLA deadline expires or customer frustration triggers the distress threshold, the ticket is elevated to 'ESCALATED'. A permanent audit record is committed to the Escalation Log.

Stage 6: Resolution, Customer Feedback & Audit Archival

Once resolved, the agent marks the ticket as 'RESOLVED'. The SLA clock stops, and the resolution timestamp is permanently recorded. Customers retain the capability to reopen tickets if the issue recurs.

5. Measurable Business Impact & Performance Metrics

• **75% Reduction in First Response Time (FRT):** AI-suggested response drafts enable agents to respond accurately in seconds.
• **Near-Zero Unnoticed SLA Breaches:** Dynamic color-coded SLA badges alert agents hours before a deadline expires.
• **Automated VIP/Urgent Escalation:** Ensures critical system crashes receive immediate executive attention without manual triage delay.